The Photoelectric Effect

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Introduction:

The photoelectric effect happens when light rays hit the surface of a metallic object, providing the electrons within the material with enough energy to be released. Under the classical theory of waves, light is an electromagnetic wave whose energy depends on its amplitude. The greater the amplitude of the light wave, the higher the intensity/brightness of the light, which means that more energy would be transferred to the electrons. This theory was later contradicted as experiments observed that the energy at which the electrons left the material was only dependent on the frequency (or colour) of the light rather than the intensity. Albert Einstein theorized that light was transmitted through small packets of energy called photons rather than electromagnetic waves, with each photon carrying energy which was proportional to the frequency of the light. The objective of this experiment is to observe the photoelectric effect and confirm Einstein's photon model of light. By separating the spectral lines emitted by a mercury vapour lamp into 5 distinct wavelengths (yellow, green, blue, blue/violet, purple), each wavelength can be shone onto a photodiode apparatus, allowing us to measure the stopping voltage (voltage needed to bring the emitted photocurrent to zero) across the first and second-order wavelengths with the use of a multimeter. This experiment aims to determine Planck's constant from the slope of the relationship between the stopping potential and frequency, identify the work function of the photodiode apparatus, and observe whether light intensity affects the stopping potential of an electron.

Theory:

Einstein's theory of light being photons represented the amount of energy in a photon by the equation $E_{Photon} = hf = \frac{hc}{\lambda}$, where h is Planck's constant ($6.636 \times 10^{-34} J \bullet s$), c is the speed of light, and λ is the wavelength of the light. The minimum energy needed for an electron to escape the surface of a metallic object is represented by the work function (ϕ) and varies by material. By applying the law of conservation of energy, the energy of a photon that is about to hit the surface of the metal is given by $E_{Photon} = K_e + \phi = hf$, where K_e is the leftover kinetic energy given to the electron. To measure the maximum kinetic energy of the electrons leaving the surface, an electric potential (ΔV_{stop}) opposing the direction of travel of the electrons is applied across the emitter and collector plates. When this potential is strong enough, it will decelerate the electrons being emitted from the metal, eventually converting all of the kinetic energy of the electron into electrical potential energy represented by $K_e = e\Delta V_{stop}$. By substituting $K_e = e\Delta V_{stop}$ into the energy equation of a photon, a linear relationship between the stopping potential and frequency of light can be observed from $\Delta V_{stop} = \frac{h}{e} f - \frac{\phi}{e}$. Plotting the measured stopping potential on a graph against frequency should then therefore yield a straight line whose slope is equal to $\frac{h}{e}$ and y-intercept is equal to $\frac{-\phi}{e}$. This should allow us to determine both the Planck's constant and work function of the photodiode apparatus in the experiment.

Procedure:

The mercury vapour light source was first powered on and allowed to heat up for approximately 5 minutes before the initial use. A digital multimeter was then turned on and set to measure $\pm$20V DC voltage, with cables connecting to the V and COM terminals. The photodiode apparatus battery was

checked by connecting the COM terminal of the multimeter to the ground terminal, while the V terminal was plugged into the "+6 MIN" terminal, ensuring that a minimum voltage reading of 6V was obtained. The V terminal was then connected to the "-6 MIN" terminal on the photodiode apparatus, ensuring that a reading lower than -6V was obtained. The setup was adjusted to obtain 5 sharp spectral lines by placing a sheet of paper near the lens assembly, then adjusting the photodiode coupling bars as well as the fine tuning knobs located on the front of the mercury vapour lamp. The apparatus when then rotated until a single spectral line entered the detector slit. The protective light shield was opened to ensure the entirety of the spectral line entered the internal detector. Once the set up was completed, the stopping voltages were recorded for the spectral line entering the apparatus with the use of the multimeter. The "PUSH TO ZERO" button was used before taking every reading of the stopping voltages to ensure a stable reading was observed. The process of adjusting the apparatus to ensure only one spectral line entered the detector was repeated for all first and second-order lines and the stopping voltages observed were recorded on a table. When observing the stopping voltages for the green and yellow spectral lines, the corresponding colour filter was attached over the photodiode apparatus to block other frequencies of light entering the apparatus.

Results and Calculations:

1) Using your measurements, plot your data in such a way that the result is a straight line. Plot two sets of data: one using the "first order" data (the 5 brighter lines closer to the center), and one using the "second order" data (the 5 dimmer lines further away from the center).

Table 1: Recorded Observations of the Stopping Voltages of First and Second-Order Spectral Lines:

Colour	**Wavelength (nm)**	**1st Order ΔV_s (V)**	**2nd Order ΔV_s (V)**
Yellow	578.0	0.68±0.05	0.61±0.05
Green	546.1	0.79±0.05	0.71±0.05
Blue	435.8	1.41±0.05	1.37±0.05
Blue/Violet	404.7	1.60±0.05	1.54±0.05
Dark Violet	365.5	1.82±0.05	1.72±0.05

To plot the data in a way that the result is a straight line, the equation $\Delta V_{stop} = \frac{h}{e}f - \frac{\phi}{e}$ is plotted where the stopping voltages are plotted on the y-axis and the frequency of light is plotted on the x-axis. To obtain the frequency of each colour of light, the equation $f = \frac{c}{\lambda}$ is used where c is the speed of light (299,792,458 m/s). An example calculation for the frequency of yellow light can be shown as:

$$f = \frac{c}{\lambda} \Rightarrow f = \frac{299{,}792{,}458\ m/s}{5.78\times10^{-7}m} \Rightarrow f = 5.187 \times 10^{14}\ Hz$$

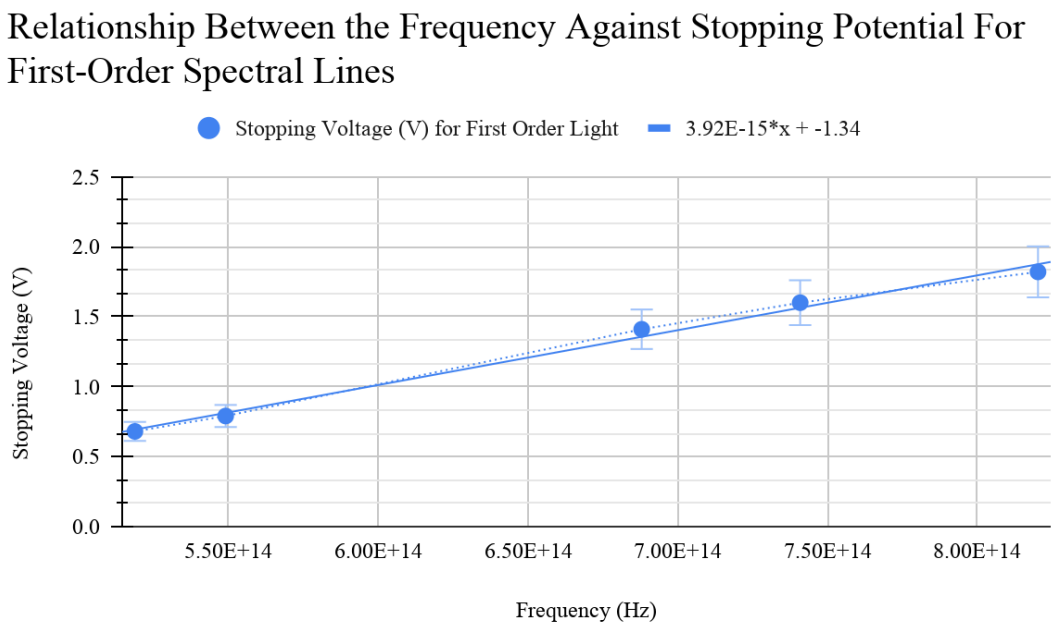


Figure 1: Relationship Between the Frequency Against Stopping Potential For First-Order Spectral Lines: From this linear relationship, a slope of 3.92 x 10^-15 is and a y-intercept of -1.34 is observed, which can be used to determine Planck's constant h and the work function ϕ.

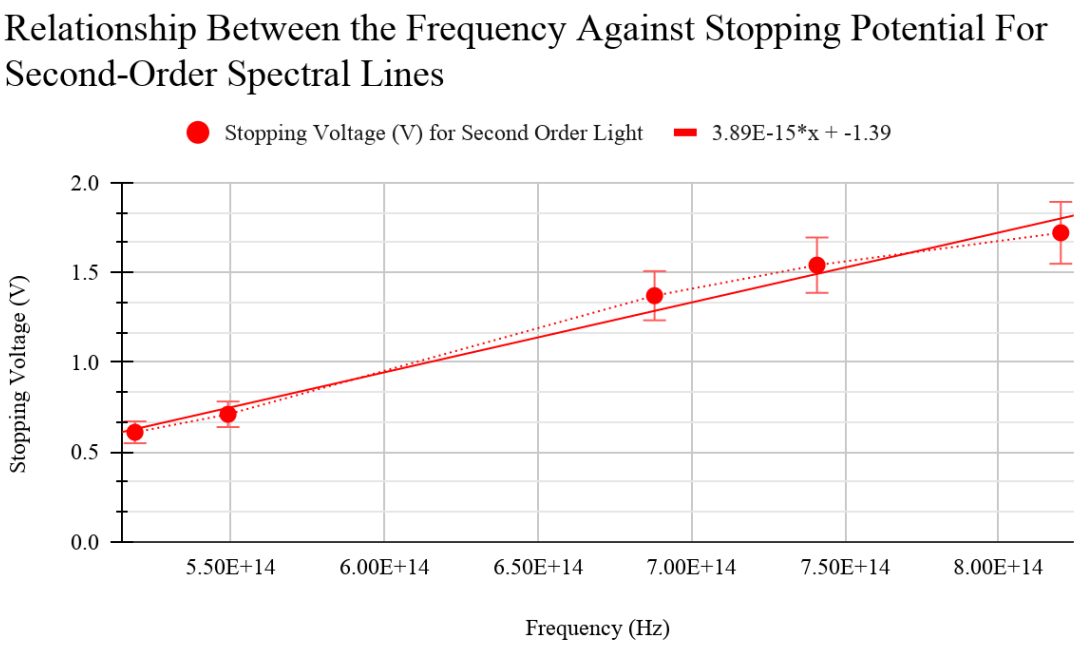


Figure 2: Relationship Between the Frequency Against Stopping Potential For Second-Order Spectral Lines: From this linear relationship, a slope of 3.89 x 10^-15 is and a y-intercept of -1.39 is observed, which can be used to determine Planck's constant h and the work function ϕ.

2) Fit your data with a line. Using the slope and the intercept for each data set, determine constant h, and the work function (in units of eV) of the photocell φ.

To calculate the Planck's constant for the first and second-order spectral lines, the slope ($m = \frac{h}{e}$) from the line of best fit's equation $\Delta V_{stop} = \frac{h}{e}f - \frac{\phi}{e}$ is rearranged to be $m \cdot e = h$. Similarly the work function ϕ is determined from the y-intercept of the linear equation given by both graphs by rearranging $-b = \frac{\phi}{e} \Rightarrow \phi = -b \cdot e$. The Planck's constants and work functions for both first and second-order spectral lines are calculated to be:

$$h_{first} = m_{first} \cdot e = (3.92 \times 10^{-15}\, V \cdot s)(1.602 \times 10^{-19}\, C) = 6.28 \times 10^{-34}\, J \cdot s$$
$$\phi_{first} = -b_{first} \cdot e = -(-1.34V)(1e) = 1.34\, eV$$

$$h_{secomd} = m_{second} \cdot e = (3.89 \times 10^{-15}\, V \cdot s)(1.602 \times 10^{-19}\, C) = 6.23 \times 10^{-34}\, J \cdot s$$
$$\phi_{second} = -b_{second} \cdot e = -(-1.39V)(1e) = 1.39\, eV$$

Table 2: Summary of Calculated Planck's Constant and Work Function Values From First and Second-Order Spectral Lines:

Variable	First-Order Observations	Second-Order Observations
Slope (m)	$3.92 \times 10^{-15}\, V \cdot s$	$3.89 \times 10^{-15}\, V \cdot s$
Y-intercept (b)	$-1.34V$	$-1.39V$
Planck's Constant (h)	$6.28 \times 10^{-34}\, J \cdot s$	$6.23 \times 10^{-34}\, J \cdot s$
Work Function (ϕ)	$1.34\, eV$	$1.39\, eV$

3) You should have two measurements of h and φ (one from the first order data, and one from the second order data). Use these two measurements to quote a final result (with uncertainty) for these two quantities

From table 2, taking the average of both Planck's constant measurements and work function values yields:

$$h_{avg} = \frac{h_{first} + h_{second}}{2} = \frac{6.28\times10^{-34} J\cdot s + 6.23\times10^{-34} J\cdot s}{2} = 6.255 \times 10^{-34}\, J \cdot s$$
$$\phi_{avg} = \frac{\phi_{first} + \phi_{second}}{2} = \frac{1.34\, eV + 1.39\, eV}{2} = 1.365\, eV$$

Using the half-range formula to find uncertainty for both values yields:

$$\delta h = \frac{|h_{first} - h_{second}|}{2} = \frac{|6.28\times10^{-34}\,J\cdot s - 6.23\times10^{-34}\,J\cdot s|}{2} = 0.03 \times 10^{-34}\,J \cdot s$$

$$\delta\phi = \frac{|\phi_{first} - \phi_{second}|}{2} = \frac{|1.34\,eV - 1.39\,eV|}{2} = 0.03\,eV$$

Therefore, the calculated final quoted measurements for Planck's constant and the work function given from the observations are:

$$h = (6.26 \times 10^{-34} \pm 0.03)\,J \cdot s$$
$$\phi = (1.37 \pm 0.03)\,eV$$

4) Compare your result for h with the accepted value including uncertainty. Is your result consistent with the accepted value (does the accepted value fall within your uncertainty range)? If not, determine the percent error by which your results differs from the expected result.

Comparing the results of Planck's constant and the work function from the experiment to the expected theoretical values of both constants, the expected Planck's constant value of $6.636 \times 10^{-34}\,J \cdot s$ does not fall within the experimental range of error, which was calculated to be from $(6.26 \times 10^{-34} \pm 0.03)\,J \cdot s$. The percent error by which the experimental results differ from the expected result can be calculated to be:

$$\%\,Error = \frac{|h_{experiment} - h_{expected}|}{h_{expected}} \times 100\% = \frac{|6.255\times10^{-34} - 6.636\times10^{-34}|}{6.636\times10^{-34}} \times 100\% = 5.74\%$$

Some sources of experimental error that yielded a 5.74% difference between the expected and experimental values of Planck's constant could be because other sources of light that were not emitted by the mercury vapour lamp entered the photodiode apparatus, affecting the stopping voltage reading on the multimeter, or because the spectral lines were slightly off-center or out of focus.

Discussion and Conclusions:

The experiment yielded a measured Planck's constant of $h = (6.26 \times 10^{-34} \pm 0.03)\,J \cdot s$ and a work function of $\phi = (1.37 \pm 0.03)\,eV$ for the photodiode apparatus. The stopping voltage was observed to increase proportionally with the frequency of the incident light across all five spectral lines, producing a clear linear relationship in both the first and second order datasets. This directly confirms Einstein's photon model prediction that the maximum kinetic energy of emitted electrons depends on the frequency of the incident light and not the intensity which is inconsistent with the classical wave model. Therefore, the experiment was completed successfully in observing the photoelectric effect and confirming Einstein's photon model.

The measured value of h does not fall within the experimental uncertainty of the accepted value $6.636 \times 10^{-34}\ J \cdot s$, corresponding to a percent error of 5.74%. . While this discrepancy places the accepted value outside the uncertainty range, an error of this size is reasonable given the sensitivity of the apparatus to alignment and stray light, small shifts in the position of the spectral line on the detector slit, or the entry of ambient light alongside the mercury lamp's emission. Therefore, the discrepancy is attributed to experimental limitations rather than a failure of the underlying photon model. Further analysis in the Appendix supports this conclusion. The stopping voltage differed only slightly between the brighter first and second order lines of the same colour, despite a substantial difference in intensity between the two orders. This result is expected under the photon model, since first and second order lines of the same colour share the same frequency and should theoretically produce identical stopping voltages regardless of brightness. The small, consistent gap observed between orders is more likely attributed to experimental factors, such as longer stabilization times for dimmer lines.

Overall, despite the moderate percent error in the measured value of Planck's constant, the experiment's results including the frequency dependence of the stopping voltage, the near-equivalence of stopping voltages across intensity levels, and the existence of a well-defined threshold frequency collectively support the photon model of light over the classical wave model.

References

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Libretexts. (2025, March 21). 5.2: Wavelength and frequency calculations. Chemistry LibreTexts. https://chem.libretexts.org/Bookshelves/Introductory_Chemistry/Introductory_Chemistry_%28CK-12%29/05%253A_Electrons_in_Atoms/5.02%253A_Wavelength_and_Frequency_Calculations

Appendix:

1) In your data set, you should have observed the stopping voltage for each color twice (once for a bright first order line, and once for a dim second order line). Was the stopping voltage significantly different for the bright line as opposed to the dim line of the same colour? Comment on your observations.

   When comparing the brighter first order incident lights to the dimmer second order datasets, we noticed a consistency that the first order dataset were all higher than the second order ones which should indicate that brightness has an effect on stopping voltage. However, once you account for measurement uncertainty, the results then all fall within range of each other. Therefore this shows that the intensity of the light does not affect the stopping voltage.

2) Does your answer to the previous question support the wave model of light, or the Einstein/Planck model of light? Explain. (Re-read the pre-lab questions if necessary)

These results support Einstein's photon model. With the classic wave model, stopping voltage would only depend on the intensity/amplitude. From our results, we saw that there was a very small difference between the first and second order voltages which is accounted for by uncertainty. Therefore since the voltages are the same with both the dim and bright light, the wave model is disproven. Furthermore, when comparing the voltages of the different frequencies, we can see that not only is there a difference in voltages, the voltages increase proportionally with the frequency of the light which follows Einstein's model.

3) The threshold frequency is the minimum frequency of light needed to release electrons from a material. Determine the threshold frequency for this apparatus. What is the associated threshold wavelength (the longest wavelength which releases electrons from the apparatus)? Quote your result with uncertainty. (You may need to review the rules for propagating uncertainty)

To determine the threshold frequency, we set $K_e = 0$. Then after rearranging the equation, we get:

$$f_{threshold} = \frac{\phi}{h} \Rightarrow f_{threshold} = \frac{(1.37eV)(1.602\times10^{-19}J/eV)}{6.26\times10^{-34}J\cdot s} \Rightarrow f_{threshold} = 3.51 \times 10^{14}\,Hz$$

For the threshold wavelength, we use the equation $\lambda = \frac{c}{f}$ which gets us:

$$\lambda_{threshold} = \frac{c}{f_{threshold}} \Rightarrow \lambda_{threshold} = \frac{299{,}792{,}458\,m/s}{3.51\times10^{14}\,Hz} \Rightarrow \lambda_{threshold} = 854\,nm$$

For uncertainty propagation, since $f_{threshold} = \frac{\phi}{h}$, relative uncertainties add up getting us:

$$\frac{\delta f}{f} = \frac{\delta\phi}{\phi} + \frac{\delta h}{h} = \frac{0.03}{1.37} + \frac{0.03\times10^{-34}}{6.26\times10^{-34}} = 0.0219 + 0.0048 = 0.0267$$
$$\delta f = f \times 0.0267 = (3.51 \times 10^{-34}) \times 0.0267 = 0.094 \times 10^{14}\,Hz$$

Since $\lambda = \frac{c}{f}$ is also a quotient (with c being exact), we can apply the uncertainty calculation:

$$\delta\lambda = \lambda \times 0.0267 = (854) \times 0.0267 = 22.8\,nm$$

Therefore our final results are:

$$f_{threshold} = (3.51 \pm 0.09) \times 10^{14}\,Hz$$
$$\lambda_{threshold} = (854 \pm 23)\,nm$$

4) Einstein hypothesized that light comes in packages of energy hf called photons. Does that mean that a bright light of a certain wavelength has the same total energy as a dim light of the same

wavelength? Using the concept of photons, explain the difference between a bright light and a dim light.

No, a bright light does not have the same total energy as a dim light of the same wavelength. Since each photon's energy depends only on wavelength $E_{Photon} = K_e + \phi = hf$, every photon in both the bright and dim light carries the exact same individual energy. The difference between a bright and a dim light of the same wavelength lies in the number of photons delivered at a time: a bright light has a much higher amount of photons than a dim light, so it has more total energy per second simply because it's made up of more individual packets of that fixed per-photon energy, not because any single photon is more energetic.